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Inventor(s)	JUNG; Hyunjun et al.

OXYGEN SATURATION MEASUREMENT GUIDE METHOD AND WEARABLE DEVICE THEREFOR

Abstract

An embodiment may include: an inertial sensor; a PPG sensor; a display module; a processor; and a memory connected electrically to the processor and configured to store instructions executable by the processor, wherein the instructions, when executed by the processor, cause the wearable device to: in response to a request for measurement of oxygen saturation, acquire motion data from the inertial sensor; responsive to a determination on the basis of the motion data that there is no motion in a wearable device, determine a wearing hand on which the wearable device is worn, on the basis of the motion data; detect a tilt of the wearable device while measuring the oxygen saturation by means of the PPG sensor; distinguish between wearing positions of the wearable device on the basis of the wearing hand and the tilt; and provide different user interfaces according to the wearing positions.

Inventors: **JUNG; Hyunjun (Suwon-si, KR), KIM; Jinho (Suwon-si, KR), JIN; Gunwoo (Suwon-si, KR)**

Applicant: **SAMSUNG ELECTRONICS CO., LTD.** (Suwon-si, KR)

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Background/Summary

BACKGROUND

[0001] An embodiment of the disclosure relates to an oxygen saturation measurement guide method and a wearable device therefor.

[0002] In accordance with development of digital technologies, various types of electronic devices, such as a mobile terminal, a personal digital assistant (PDA), an electronic notebook, a smartphone, a tablet personal computer (PC), or a wearable device, have been widely used. Electronic devices have been continuously improved in terms of hardware and/or software of the electronic devices to support and increase functions thereof.

[0003] For example, an electronic device like a wearable device may be provided in various forms, such as a smartwatch, smart glasses, and smart bands that may come in contact with (or worn on) the user's body. A wearable device may collect and analyze various information (e.g., biometric or activity information) associated with a user so as to provide various functions (e.g., exercise information or health information) to the user.

[0004] For example, a wearable device may measure oxygen saturation using a pulse oximetry method. Oxygen saturation may an indicator used in a vital check along with electrocardiogram, blood pressure, pulse, respiratory rate, and body temperature. A pulse oximetry method may be a method of measuring oxygen saturation using the ratio of the absorbance of increased blood flow at two wavelengths (e.g., RED, Infrared) using temporary volume changes in arterial blood caused by cardiac output.

SUMMARY

[0005] According to an embodiment of the disclosure, a wearable device may include an inertial sensor, a PPG sensor, a display module, a processor operatively coupled to at least one of the inertial sensor, the PPG sensor, and the display module, and a memory connected electrically to the processor and configured to store instructions executable by the processor, wherein the instructions, when executed by the processor, cause the wearable device to, in response to a request for measurement of oxygen saturation, acquire motion data from the inertial sensor, responsive to a determination, based on the motion data, that there is no motion of the wearable device, determine a wearing hand on which the wearable device is worn, based on the motion data, detect a tilt of the wearable device during the measurement of oxygen saturation by means of the PPG sensor, classify wearing positions of the wearable device, based on the wearing hand and the tilt, and provide different user interfaces according to the wearing positions.

[0006] According to an embodiment, an operation method of a wearable device may include an operation of, in response to a request for measurement of oxygen saturation, acquiring motion data from an inertial sensor included in the wearable device, an operation of, if it is determined, based on the motion data, that there is no motion of a wearable device, determining a wearing hand on which the wearable device is worn, based on the motion data, an operation of detecting a tilt of the wearable device during the measurement of oxygen saturation by means of a PPG sensor included in the wearable device, an operation of classifying wearing positions of the wearable, device based on the wearing hand and the tilt, and an operation of providing different user interfaces according to the wearing positions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a block view illustrating an electronic device in a network environment according to an embodiment.

[0008] FIG. **2** is a front perspective view illustrating a wearable device according to various embodiments.

[0009] FIG. **3** is a rear perspective view illustrating a wearable device according to various embodiments.

[0010] FIG. **4** is a flowchart illustrating an operation method of a wearable device according to an embodiment.

[0011] FIGS. **5**A to **5**D are views illustrating an example of a user interface provided in a wearable device according to an embodiment.

[0012] FIG. **6** is a flowchart illustrating a method of classifying wearing positions according to a wearing hand in a wearable device according to an embodiment.

[0013] FIGS. **7**A and **7**B are views illustrating an example of classifying wearing positions according to a wearing hand in a wearable device according to an embodiment.

[0014] FIG. **8** is a flowchart illustrating a method of providing a user interface depending on wearing suitability in a wearable device according to an embodiment.

[0015] FIGS. **9**A and **9**B are views illustrating an example of classifying wearing suitability of a wearable device according to an embodiment.

[0016] FIG. **10** is a view illustrating an example of providing a guide associated with oxygen saturation measurement in a wearable device according to an embodiment.

DETAILED DESCRIPTION

[0017] FIG. **1** is a block diagram illustrating an electronic device **101** in a network environment **100** according to certain embodiments.

[0018] Referring to FIG. **1**, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0019] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation.

According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.
[0020] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.
[0021] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.
[0022] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.
[0023] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).
[0024] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.
[0025] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram

device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0026] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0027] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0028] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0029] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

[0030] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0031] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0032] The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0033] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0034] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module).

A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™ wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a **5**th generation (5G) network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.
[0035] The wireless communication module **192** may support a 5G network, after a 4th generation (4G) network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.
[0036] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.
[0037] According to certain embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, an RFIC disposed on a first surface (e.g., the bottom surface) of the PCB, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the PCB, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.
[0038] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0039] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an Internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0040] The electronic device according to various embodiments disclosed herein may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smart phone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. The electronic device according to embodiments of the disclosure is not limited to those described above.

[0041] It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or alternatives for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to designate similar or relevant elements. A singular form of a noun corresponding to an item may include one or more of the items, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “a first”, “a second”, “the first”, and “the second” may be used to simply distinguish a corresponding element from another, and does not limit the elements in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with/to” or “connected with/to” another element (e.g., a second element), it means that the element may be coupled/connected with/to the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0042] As used herein, the term “module” may include a unit implemented in hardware, software, or firmware, and may be interchangeably used with other terms, for example, “logic,” “logic block,” “component,” or “circuit”. The “module” may be a minimum unit of a single integrated component adapted to perform one or more functions, or a part thereof. For example, according to an embodiment, the “module” may be implemented in the form of an application-specific integrated circuit (ASIC).

[0043] Various embodiments as set forth herein may be implemented as software (e.g., the program

140) including one or more instructions that are stored in a storage medium (e.g., the internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a complier or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0044] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., Play Store™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0045] According to various embodiments, each element (e.g., a module or a program) of the above-described elements may include a single entity or multiple entities, and some of the multiple entities mat be separately disposed in any other element. According to various embodiments, one or more of the above-described elements may be omitted, or one or more other elements may be added. Alternatively or additionally, a plurality of elements (e.g., modules or programs) may be integrated into a single element. In such a case, according to various embodiments, the integrated element may still perform one or more functions of each of the plurality of elements in the same or similar manner as they are performed by a corresponding one of the plurality of elements before the integration. According to various embodiments, operations performed by the module, the program, or another element may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0046] To improve the accuracy of pulse oximetry, methods such as applying pressure to reduce venous blood components, guiding the user to change the measurement position (e.g., below or above the elbow), and using signal processing techniques to remove noise may be used. However, it may be difficult to apply the methods above to a wearable device.

[0047] An embodiment may provide a method and a device which determine a wearing position by means of an inertial sensor of a wearable device and provide a user guide corresponding to the determined wearing position or determine a quality of oxygen saturation measured through a photoplethysmography (PPG) sensor and provide a user guide corresponding to the determined quality of oxygen saturation.

[0048] According to an embodiment, by determining a wearing position by means of an inertial sensor of a wearable device and providing a user guide corresponding to the determined wearing position, the user may be guided to a precise position in which the user wears the wearable device to measure oxygen saturation.

[0049] According to an embodiment, by determining the quality of oxygen saturation measured through a PPG sensor and providing a user guide corresponding to the determined quality of the oxygen saturation, the accuracy of the oxygen saturation measurement may be improved.

[0050] According to an embodiment, by providing different user guides depending on the wearing position or the quality of oxygen saturation, the user may follow the guides to improve a wearing state of the wearable device and increase the usability of the wearable device.

[0051] According to an embodiment, by providing gradually different guides based on motion detection or the quality of oxygen saturation, each user may be guided to an optimized measurement position to increase the success rate of the measurement.

[0052] FIG. **2** is a front perspective view illustrating a wearable device according to various embodiments, and FIG. **3** is a rear perspective view illustrating a wearable device according to various embodiments.

[0053] Referring to FIGS. **2** and **3**, the wearable device (e.g., the electronic device **101** in FIG. **1**) according to various embodiments may include a housing **210** including a first surface (or front surface) **210**A, a second surface (or rear surface) **210**B, and a lateral surface **210**C surrounding a space between the first surface **210**A and the second surface **210**B, and a coupling member **250** or **260** connected to at least a portion of the housing **210** and configured to detachably couple the wearable device **200** to the body (e.g., a wrist, an ankle of the like) of a user. According to another embodiment (not shown), the housing may refer to a structure for configuring a portion of the first surface **210**A, the second surface **210**B, and the lateral surface **210**C in FIG. **2**.

[0054] According to an embodiment, at least a portion of the first surface **210**A may be made of a substantially transparent front plate **201** (e.g., a glass plate including various coating layers or polymer plate). The second surface **210**B may be made of a substantially opaque rear plate **207**. The rear plate **207** may include, for example, coated or colored glass, ceramic, polymers, metals (e.g., aluminum, stainless steel (STS), or magnesium), or a combination of at least two thereof. The lateral surface **210**C may be coupled to the front plate **201** and the rear plate **207** and made of a lateral bezel structure (or "lateral member") **206** including a metal and/or polymer. In an embodiment, the rear plate **207** and the lateral bezel structure **206** may be integrally configured and include an identical material (e.g., a metal material such as aluminum). The coupling member **250** or **260** may be made of various materials and in various shapes. The coupling member may be configured to be integrally and to multiple unit links to be movable with each other by using a woven fabric, leather, rubber, urethane, metal, ceramic, or a combination of at least two of the materials.

[0055] According to various embodiments, the wearable device **200** may include at least one of a display **220** (e.g., the display module **160** in FIG. **1**), an audio module **205** or **208**, a sensor module **211** (e.g., the sensor module **176** in FIG. **1**), a key input device **202**, **203**, or **204** (e.g., the input module **150** in FIG. **1**), and a connector hole **209**. In an embodiment, the wearable device **200** may omit at least one of the components (e.g., the key input device **202**, **203**, or **204**, the connector hole **209**, or the sensor module **211**) or additionally include another component.

[0056] The display **220** may be exposed to outside through, for example, a substantial portion of the front plate **201**. The display **220** may have a shape corresponding to the shape of the front plate **201** and have various shapes such as a circle, an oval, or a polygon. The display **220** may be combined to or disposed adjacent to a touch sensing circuit, a pressure sensor for measuring a strength (pressure) of a touch, and/or a fingerprint sensor.

[0057] The audio module **205** or **208** may include a microphone hole **205***a* and a speaker hole **208**. A microphone configured to acquire a sound from outside may be disposed in the microphone hole **205***a* and in an embodiment, multiple microphones may be arranged to detect a direction of a sound. The speaker hole **208** may be used for an outer speaker and a receiver for calling.

[0058] The sensor module **211** may generate an electrical signal or a data value corresponding to an internal operation state or external environment state of the wearable device **200**. The sensor module **211** may include, for example, a biosensor module **211** (e.g., a HRM sensor) disposed on the second surface **210**B of the housing **210**. The wearable device **200** may further include at least one sensor module not shown in the drawings, for example, a gesture sensor, a gyro sensor, an air pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared

(IR) sensor, a biometric sensor, a temperature sensor, humidity sensor, or an illuminance sensor.
[0059] The key input device **202**, **203**, or **204** may include a wheel key **202** disposed on the first surface **210**A of the housing **210** and rotatable in at least one direction, and/or a side button key **202** or **203** disposed at the lateral surface **210**C of the housing **210**. The wheel key may have a shape corresponding to the front plate **202**. In another embodiment, the wearable device **200** may not include a portion or entirety of the key input device **202**, **203**, or **204**, and the excluded key input device **202**, **203**, or **204** may be implemented in various forms such as a soft key on the display **220**. The connector hole **209** may include another connector hole (not shown) capable of receiving a connector (e.g., a USB connector) configured to transmit or receive power and/or data to or from an external electronic device and a connector for transmitting or receiving an audio signal to or from an external electronic device. The wearable device **200** may further include, for example, a connector cover (not shown) configured to cover a portion of the connector hole **209** and block the ingress of external foreign substances into the connector hole.
[0060] The coupling member **250** or **260** may be detachably coupled to at least a portion of the housing **210** by using a locking member **251** and **261**. The coupling member **250** or **260** may include one or more of a fixation member **252**, a fixation member fastening hole **253**, a band guide member **254**, and a band fixation ring **255**.
[0061] The fixation member **252** may be configured to fix the coupling member **250** or **260** and the housing **210** to a body portion (e.g.: wrist and ankle) of a user. The fixation member fastening hole **253** may fix the coupling member **250** or **260** and the housing **210** to a body portion of a user by counteracting with the fixation member **252**. The band guide member **254** is configured to limit the motion range of the fixation member **252** when the fixation member **252** is fastened to the fixation member fastening hole **253** so that the coupling member **250** or **260** is closely bound to a body portion of a user. The band fixation ring **255** may limit the motion range of the coupling member **250** or **260** in a state where the fixation member **252** is fastened to the fixation member fastening hole **253**.
[0062] According to an embodiment of the disclosure, a wearable device **200** may include an inertial sensor (e.g., sensor module **176**), a PPG sensor (e.g., sensor module **176**), a display module **220**, a memory **130**, and a processor **120** operatively connected to at least one of the inertial sensor, the PPG sensor, the display module, or the memory, wherein the processor may be configured to, in response to a request for measurement of oxygen saturation, acquire motion data from the inertial sensor, when it is determined, based on the motion data, that there is no motion of the wearable device, determine a wearing hand on which the wearable device is worn, detect a tilt of the wearable device while measuring the oxygen saturation by means of the PPG sensor, classify wearing positions of the wearable device based on the wearing hand and the tilt, and provide different user interfaces according to the wearing positions.
[0063] The processor may be configured to, in case that the wearing hand corresponds to the left hand, apply a left-hand tilt detection algorithm and, in case that the wearing hand corresponds to the right hand, apply a right-hand tilt detection algorithm.
[0064] The processor may be configured to determine whether an angle between a wrist direction of the wearable device and the ground exceeds a first reference value according to the left-hand tilt detection algorithm and classify the wearing position of the wearable device according to a result of the determination.
[0065] The processor may be configured to determine whether the angle between the wrist direction of the wearable device and the ground is less than a second reference value according to the right-hand tilt detection algorithm and classify the wearing position of the wearable device according to a result of the determination.
[0066] The processor may be configured to, in case that the wearing position corresponds to a first position, provide a first user interface associated with the first position, in case that the wearing position corresponds to a second position, provide a second user interface associated with the

second position, and in case that the wearing position corresponds to a third position, provide a third user interface associated with the third position, wherein the first position corresponds to a case where the angle between the wrist direction of the wearable device and the ground has a value exceeding a first reference value or less than a second reference value, the second position corresponds to a case where the angle between the wrist direction of the wearable device and the ground has a value between the first reference value and the second reference value, and the third position corresponds to a case where the angle between the wrist direction of the wearable device and the ground has a value opposite to the first position.
[0067] The processor may be configured to, in case that the wearing position corresponds to the first position or the second position, maintain oxygen saturation measurement and, in case that the wearing position corresponds to the third position, stop oxygen saturation measurement.
[0068] The processor may be configured to acquire an amplitude of alternating current (AC) and an amplitude of direct current (DC) from a PPG signal acquired from the PPG sensor, in case that the amplitude of the AC exceeds the amplitude of the DC, determine that a wearing state of the wearable device is suitable, and in case that the amplitude of the AC is equal to or less than the amplitude of the DC, determine that the wearing state of the wearable device is unsuitable.
[0069] The processor may be configured to, when it is determined that the wearing state of the wearable device is suitable, maintain oxygen saturation measurement.
[0070] The processor may be configured to determine whether the measured oxygen saturation exceeds a reference value, in case that the measured oxygen saturation exceeds the reference value, provide an oxygen saturation measurement value, and in case that the measured oxygen saturation is equal to or less than the reference value, provide the oxygen saturation measurement value and a warning notification.
[0071] The processor may be configured to, when it is determined that the wearing state of the wearable device is unsuitable, stop oxygen saturation measurement.
[0072] The processor may be configured to, in case that the number of oxygen saturation measurement stops corresponds to a designated number of times, provide a first measurement guide, and in case that the number of oxygen saturation measurement stops does not correspond to the designated number of times, provide a second measurement guide.
[0073] The processor may be configured to determine whether there is no motion of the wearable device after providing the first measurement guide or the second measurement guide.
[0074] The processor may be configured to, after the oxygen saturation measurement is requested, in case that the number of detections of the motion of the wearable device corresponds to a designated number of detections, provide a first guide, and after the oxygen saturation measurement is requested, in case that the number of detections of the motion of the wearable device does not correspond to the designated number of detections, provide a second guide.
[0075] FIG. **4** is a flowchart **400** illustrating an operation method of a wearable device according to an embodiment.
[0076] Referring to FIG. **4**, in operation **401**, a processor (e.g., the processor **120** in FIG. **1**) of the wearable device (e.g., the electronic device **101** in FIG. **1** or the wearable device **200** in FIG. **2**) according to an embodiment may detect a motion. For example, the processor **120** may detect whether there is a motion of the wearable device **200** by using a sensor module (e.g., the sensor module **176** in FIG. **1**) included in the wearable device **200**. For example, the sensor module **176** may correspond to an inertial sensor, and the inertial sensor may include an acceleration sensor and a gyro sensor. According to one or more embodiments, the inertial sensor is always operated and thus the processor **120** may acquire motion data from the inertial sensor in real time. The processor **120** may, in case that a measurement of oxygen saturation (e.g., SpO2) is requested, detect whether there is a motion of the wearable device **200**. In case that a motion is detected when measuring oxygen saturation, it may cause problems with the oxygen saturation measurement. The processor **120** may, in case that a motion is not detected from the wearable device **200**, start oxygen

saturation measurement in order to prevent mismeasurement of oxygen saturation.
[0077] According to an embodiment, the processor **120** may convert the acquired motion data (e.g., a sensing value of x, y, and z) into an Euler angle, and detect a motion of the wearable device **200** through whether a change (e.g., a motion of 5 degrees or more in 3 seconds) in the converted Euler angle (e.g., one of roll, pitch, or yaw) greater than or equal to a determined threshold is detected during a determined time period. Alternatively, the processor **120** may calculate the amount of change in x, y, and z by using the value of the acquired motion data itself, and detect a motion of the wearable device **200** through whether the calculated amount of change dictates the detection of change greater than or equal to a determined threshold. There are various methods to detect the motion of the wearable device **200**, and the processor **120** may detect the motion by using various known calculation methods.
[0078] In operation **403**, the processor **120** may detect (or determine) a wearing hand of the wearable device **200**. The processor **120** may, in case that the motion of the wearable device **200** has not been detected, detect a wearing hand of the wearable device **200**, based on the motion data acquired during a process in which the user is wearing the wearable device **200** and raises a hand for measuring oxygen saturation. Alternatively, the processor **120** may provide a user interface for querying the user whether the wearable device **200** is worn on the left or right hand and acquire information about the wearing hand from the user. Alternatively, the processor **120** may acquire information about the wearing hand of the wearable device **200** from configuration information of the wearable device **200**. The configuration information may be acquired by directly inputting information about the wearing hand of the wearable device **200**.
[0079] In operation **405**, the processor **120** may detect a tilt, based on the motion data. The processor **120** may, based on the converted Euler angle, detect the tilt of the wearable device **200**. The reason for detecting the wearing hand of the wearable device **200** first, and then detecting the tilt of the wearable device **200**, may be that the direction of the wearable device **200** alone does not indicate whether a current position of the wearable device **200** is higher or lower than the user's heart. For example, this may be to distinguish when the user is wearing the wearable device **200** on the left wrist and is raising the left hand, and when the user is wearing the wearable device **200** on the right wrist and is lowering the right wrist.
[0080] According to an embodiment, the processor **120** may, in case that the wearing hand has been determined, start oxygen saturation measurement by means of a photoplethysmography (PPG) sensor (e.g., the sensor module **176** in FIG. **1**). Detecting the wearing hand may be determined before the oxygen saturation measurement. The sensor module **176** may further include a PPG sensor. The PPG sensor may include a light-emitting unit (e.g., an LED (e.g., RED or infrared)), a light-receiving unit (e.g., a photo diode (PD)), an optical structure configured to acquire reflective PPG, and a signal processing unit configured to process the acquired PPG signal. For example, the light-emitting unit and the light-receiving unit of the PPG sensor may be configured as an array of LEDs and PDs having multiple wavelengths, and a spectrometer light source outputting a multi-wavelength laser may be used. The signal processing unit may calculate oxygen saturation (or an oxygen saturation value) by using conventionally well-known method (e.g., measuring an amount of light output from the light-emitting unit that transmits through the user's body (e.g., a finger or wrist) or is reflected and reaches the light-receiving unit). For example, the signal processing unit may calculate oxygen saturation by using a ratio of AC to DC components for each wavelength included in the PPG signal.
[0081] For example, the DC component is derived from substances that produce a constant reflection, such as skin, muscle and bone, and venous blood. In case that the human body is stationary and has less motion, the AC component is made mainly of light reflected from the pulsation of arterial blood. The AC component may vary depending on the heart rate and arterial thickness. More light may be reflected or transmitted during systole than during diastole. During systole, arterial blood pressure may increase because the heart is pumping out blood. Increased

blood pressure may cause arteries to expand and increase blood flow in the arteries, and increased blood flow may increase light absorption. During diastole, the blood pressure may decrease, and light absorption may also decrease. The processor **120** may apply a different tilt detection algorithm depending on the wearing hand of the wearable device **200** while measuring oxygen saturation.

[0082] In case that the wearing hand of the wearable device **200** is the left hand, the pitch, which is a measure of the degree to which the angle between the wrist direction of the wearable device **200** and the ground is tilted, may have a determined positive value (e.g., $\theta>60°$, exceeding a first threshold). On the contrary, in case that the wearing hand of the wearable device **200** is the right hand, the pitch, which is a measure of the degree to which the angle between the wrist direction of the wearable device **200** and the ground is tilted, may have a determined negative value (e.g., $\theta<-60°$, less than a second threshold). The processor **120** may apply a different tilt detection algorithm depending on the wearing hand of the wearable device **200** so as to determine a wearing position of the wearable device **200**. The processor **120** may apply a left-hand tilt detection algorithm in case that the wearing hand of the wearable device **200** is the left hand. The processor **120** may apply a right-hand tilt detection algorithm in case that the wearing hand of the wearable device **200** is the right hand.

[0083] In operation **407**, the processor **120** may determine a wearing position according to the tilt. The wearing position may refer to a position in which the user wears the wearable device **200**. The processor **120** may determine the wearing position of the wearable device **200**, based on the wearing hand of the wearable device **200** and the tilt of the wearable device **200**. The processor **120** may, in case that the wearing hand of the wearable device **200** is the left hand, determine the wearing position according to the left-hand tilt detection algorithm. The processor **120** may, in case that the wearing hand of the wearable device **200** is the right hand, determine the wearing position according to the right-hand tilt detection algorithm.

[0084] In operation **409**, the processor **120** may provide a user interfaces corresponding to the wearing position. The user interface may include at least one of text, an image, audio, or a video. The processor **120** may display the user interface through the display **220**. Alternatively, the processor **120** may output a voice related to the user interface through the audio module **205** or **208**. The processor **120** may measure oxygen saturation while performing operation **405** to operation **409**.

[0085] According to an embodiment, the processor **120** may, in case that the wearing position corresponds to a first position, provide a first user interface related to the first position. The first position refers to a wearing position of the wearable device **200** that is above the height of the user's heart and is less affected by venous blood and may correspond to a wearing guide position. The first position may correspond to a case where the angle (e.g., θ) between the wrist direction of the wearable device **200** and the ground has a value exceeding a first reference value (e.g., when the wearable device **200** is worn on the left hand, $\theta>60°$) or less than a second reference value (e.g., (e.g., when the wearable device **200** is worn on the right hand, $\theta<-60°$). The first user interface may indicate that oxygen saturation is being measured.

[0086] Alternatively, the processor **120** may, in case that the wearing position corresponds to a second position, provide a second user interface related to the second position. The second position may not correspond to the wearing guide position and refer to as a position in which the accuracy of the measurement of oxygen saturation may vary depending on wearing states of the wearable device **200**. The second position may correspond to a case where the angle (e.g., θ) between the wrist direction of the wearable device **200** and the ground has a value falling within a reference range (e.g., a value between the first reference value and the second reference value, $-60°\leq\theta\leq60°$). The second user interface may include a guide indicating that the accuracy of measurement of oxygen saturation is reduced and prompting correct wearing of wearable device **200**.

[0087] Alternatively, the processor **120** may, in case that the wearing position corresponds to a third position, provide a third user interface related to the third position. The third position may not correspond to the wearing guide position and correspond to a position that may cause problems with oxygen saturation measurement (e.g., an inaccurate value of oxygen saturation). The third position may correspond to a case where the angle (e.g., θ) between the wrist direction of the wearable device **200** and the ground has a value opposite to the first position (e.g., when the wearable device **200** is worn on the left hand, $\theta<-60°$, and when the wearable device **200** is worn on the right hand, $\theta>60°$). The third user interface may include a guide indicating that the measurement of oxygen saturation is impossible and prompting correct wearing of wearable device **200**.

[0088] Accordingly, the processor **120** may selectively maintain the oxygen saturation measurement only in case that the wearing position of the wearable device **200** corresponds to the first position or the second position. The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the first position or the second position, perform operations in FIG. **8**. Since oxygen saturation is more likely to be mismeasured when wearing position of the wearable device **200** is the third position, the processor **120** may stop measuring oxygen saturation in case that the wearing position of the wearable device **200** is the third position. The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the third position, not perform operations in FIG. **8**. The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the third position, stop measuring oxygen saturation to prevent degradation of oxygen saturation measurement quality. The processor **120** may prevent a situation that results in low quality oxygen saturation measurement, in advance.

[0089] According to an embodiment, the processor **120** may, in case that a motion of the wearable device **200** is detected before measuring oxygen saturation, determine whether the motion detection is a first occurrence. The processor **120** may provide different guides depending on whether it is the first occurrence (or more than a designated number of times (e.g., three times)). Although it is described below as providing different guides depending on whether the motion detection is a first occurrence, it is possible to provide different guides depending on whether the motion detection has occurred more than a designated number of times (e.g., three times). This is merely an implementation issue, and the disclosure is not limited thereto.

[0090] In case that the motion detection is the first occurrence, the processor **120** may provide the first guide. The first guide may include a message, such as “Do not move.” The first guide may include at least one of text, an image, audio, or a video. The processor **120** may, in case that the motion detection is the first occurrence, provide the first user interface **510** of FIG. **5**A. The processor **120** may display the first guide through a display (e.g., the display module **160** in FIG. **1** or the display **220** in FIG. **2**) of the wearable device **200** or output the first guide using a voice through a speaker (e.g., the sound output module **155** in FIG. **1** or the audio module **205** or **208** in FIG. **2**) of the wearable device **200**.

[0091] In case that the motion detection is not the first occurrence, the processor **120** may provide a second guide. The second guide may be different from the first guide. The second guide may include a message, such as “Take a stable position,” together with a position example. The second guide may include at least one of text, an image, audio, or a video. The processor **120** may, in case that the motion detection is not the first occurrence, provide the second user interface **515** of FIG. **5**A. The processor **120** may display the second guide through the display **220** or output the second guide using a voice through the audio module **205** or **208**.

[0092] FIGS. **5**A and **5**D are views illustrating an example of a user interface provided in a wearable device according to an embodiment.

[0093] FIG. **5**A is a view illustrating a user interface provided when a motion of the wearable device is detected.

[0094] Referring to FIG. **5**A, a processor (e.g., the processor **120** in FIG. **1**) of the wearable device (e.g., the electronic device **101** in FIG. **1** or the wearable device **200** in FIG. **2**) according to an embodiment may provide a first user interface **510** in case that the motion of the wearable device

200 is the first occurrence. The first user interface **510** may include the first guide such as "Do not move." The processor **120** may, in case that the motion detection of the wearable device **200** is not the first occurrence, provide a second user interface **515** of FIG. **5**A. The second user interface **515** may include a second guide such as "Take a stable position." or "Rest your elbow on a table and keep your wrist close to your heart." together with a position example. The first user interface **510** or the second user interface **515** may include at least one of text, an image, audio, or a video.

[0095] FIG. **5**B is a view illustrating a user interface provided according to a wearing position of the wearable device.

[0096] Referring to FIG. **5**B, the processor **120** may, in case that the wearing position of the wearable device **200** corresponds to a second position, provide a third user interface **520**. The third user interface **520** may include a third guide such as "Keep your arm above the heart position." The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the second position or the third position, provide a fourth user interface **525**. The fourth user interface **525** may include a message, such as "Hold the shoulder of the arm not wearing the watch (e.g., the wearable device **200**) with the hand wearing the watch," together with a position example. The third user interface **520** or the fourth user interface **525** may include at least one of text, an image, audio, or a video. The example of the third guide or the fourth guide is for illustrative purposes of the disclosure and may include different images or text depending on the wearing position.

[0097] FIG. **5**C is a view illustrating a user interface related to oxygen saturation measurement in the wearable device.

[0098] Referring to FIG. **5**C, the processor **120** may provide different user interfaces depending on the measured oxygen saturation. The processor **120** may, in case that the measured oxygen saturation corresponds to a normal value, provide a fifth user interface **530** including the measured oxygen saturation. The fifth user interface **530** may include the measured oxygen saturation and a tag button. The processor **120** may, in case that the measured oxygen saturation does not correspond to a normal value, provide a sixth user interface **535** including the measured oxygen saturation and a warning notification. The sixth user interface **535** may include the measured oxygen saturation and a warning notification (e.g., "Your blood oxygen is low. Take another measurement. If this result repeats, contact a doctor."). The fifth user interface **530** or the sixth user interface **535** may include at least one of text, an image, audio, or a video.

[0099] FIG. **5**D is a view illustrating a user interface related to oxygen saturation measurement in the wearable device.

[0100] Referring to FIG. **5**D, the processor **120** may, in case that the quality of oxygen saturation is low (e.g., wearing unsuitability), provide different user interfaces. The processor **120** may, in case that low oxygen saturation quality is first detected, may provide a seventh user interface **540**, such as "Place the watch on your wrist and fasten the strap." The seventh user interface **540** may include information on a method for correct wearing, together with an example of correct wearing. The processor **120** may, in case that low oxygen saturation quality is not first detected, provide an eighth user interface **545**. The eighth user interface **545** may include a guide such as "Remove and reattach the device" or "Reattachment detected. Remeasure oxygen saturation?" or "Reattachment detected. Start to remeasure?". The seventh user interface **540** or the eighth user interface **545** may include at least one of text, an image, audio, or a video.

[0101] The first user interface **510** to the eighth user interface **545** are examples provided to aid in understanding the disclosure and should not be construed as limiting the scope of the disclosure.

[0102] FIG. **6** is a flowchart **600** illustrating a method of classifying wearing positions according to a wearing hand in a wearable device according to an embodiment. Operations in FIG. **6** may embody operations in FIG. **4**.

[0103] Referring to FIG. **6**, in operation **601**, a processor (e.g., the processor **120** in FIG. **1**) of the wearable device (e.g., the electronic device **101** in FIG. **1** or the wearable device **200** in FIG. **2**) according to an embodiment may receive a request for oxygen saturation measurement. The user may wear the wearable device **200** and then request the oxygen saturation measurement. For example, the processor **120** may receive, from the user, receive selecting (e.g., touching) an oxygen saturation measurement button from a measurement menu or a voice input such as “Measure oxygen saturation.”

[0104] In operation **603**, the processor **120** may detect whether a motion is less than or equal to a threshold value. The processor **120** may detect whether there is no motion of the wearable device **200** by using a sensor module (e.g., the sensor module **176** in FIG. **1**) included in the wearable device **200**. For example, the sensor module **176** may correspond to an inertial sensor, and the inertial sensor may include an acceleration sensor and a gyro sensor. According to one or more embodiments, the inertial sensor is always operated and thus the processor **120** may acquire motion data from the inertial sensor in real time. The processor **120** may detect whether there is a motion of the wearable device **200**, based on the motion data. Detecting whether a motion is less than or equal to a threshold value may correspond to identifying whether there is no motion. The processor **120** may, in case that the motion is less than or equal to the threshold value, perform operation **605**.

[0105] In operation **605**, the processor **120** may determine a wearing hand of the wearable device **200**. The processor **120** may, in case that the motion of the wearable device **200** has not been detected, detect a wearing hand of the wearable device **200**, based on the motion data acquired during a process in which the user raises a hand for measuring oxygen saturation. Alternatively, the processor **120** may provide a user interface for querying the user whether the wearable device **200** is worn on the left or right hand and acquire information about the wearing hand from the user. Alternatively, the processor **120** may acquire information about the wearing hand of the wearable device **200** from configuration information of the wearable device **200**. The configuration information may be acquired by directly inputting information about the wearing hand of the wearable device **200**.

[0106] In operation **607**, the processor **120** may determine whether the wearing hand of the wearable device **200** is the left hand. The drawing shows the determination of whether the wearing hand is the left hand, but it is also possible to determine whether the wearing hand is the right hand. In general, since the user may be more likely to wear the wearable device on the left hand, it may determine whether the wearing hand is the left hand. The processor **120** may perform operation **609** in case that the wearing hand of the wearable device **200** is the left hand, and perform operation **611** in case that the wearing hand of the wearable device **200** is the right hand.

[0107] In case that the wearing hand of the wearable device **200** is the left hand, in operation **609**, the processor **120** may measure oxygen saturation and apply the left-hand tilt detection algorithm. This may be because the direction of the wearable device **200** alone does not indicate whether a current position of the wearable device **200** is higher or lower than the user's heart. For example, this may be to distinguish when the user is wearing the wearable device **200** on the left wrist and is raising the left hand, and when the user is wearing the wearable device **200** on the right wrist and is lowering the right wrist. For example, in case that the wearing hand of the wearable device **200** is the left hand, the angle (e.g., θ) between the wrist direction of the wearable device **200** and the ground may correspond to a determined positive value (e.g., θ>60°, exceeding a first reference value). Accordingly, the processor **120** may apply a different tilt detection algorithm depending on the wearing hand of the wearable device **200**.

[0108] According to an embodiment, the processor **120** may start oxygen saturation measurement by means of the PPG sensor (e.g., the sensor module **176** in FIG. **1**). The sensor module **176** may further include a PPG sensor. The processor **120** may acquire a PPG signal from the PPG sensor and calculate oxygen saturation by using a ratio of AC to DC components for each wavelength included in the acquired PPG signal. The method for calculating oxygen saturation is well known in the art and will not be described in detail.

[0109] In case that the wearing hand of the wearable device **200** is the right hand, in operation **611**, the processor **120** may measure oxygen saturation and apply the right-hand tilt detection algorithm.

In case that the wearing hand of the wearable device **200** is the right hand, the angle (e.g., θ) between the wrist direction of the wearable device **200** and the ground may correspond to a determined negative value (e.g., θ<−60°, less than a second reference value). The processor **120** may perform operation **611** and then perform operation **613**.
[0110] In operation **613**, the processor **120** may a tilt of the wearable device **200**. The processor **120** may, in case that the wearing hand of the wearable device **200** is the left hand, apply the left-hand tilt detection algorithm to detect the tilt of the wearable device **200**. The processor **120** may, in case that the wearing hand of the wearable device **200** is the right hand, apply the right-hand tilt detection algorithm to detect the tilt of the wearable device **200**.
[0111] In operation **615**, the processor **120** may determine a wearing position according to the wearing hand and the tilt. The wearing position may refer to a position in which the user wears the wearable device **200**. The processor **120** may determine the wearing position of the wearable device **200**, based on the wearing hand of the wearable device **200** and the tilt of the wearable device **200**. The processor **120** may, in case that the wearing hand of the wearable device **200** is the left hand, determine the wearing position according to the left-hand tilt detection algorithm. The processor **120** may, in case that the wearing hand of the wearable device **200** is the right hand, determine the wearing position according to the right-hand tilt detection algorithm.
[0112] In operation **615**, the processor **120** may provide a different wearing guide depending on the wearing position of the wearable device **200**. The wearing guide may include at least one of text, an image, audio, or a video. The processor **120** may display the wearing guide through the display **220** or output a voice related to the wearing guide through the audio module **205** or **208**. The processor **120** may measure oxygen saturation while performing operation **609** to operation **617**.
[0113] According to an embodiment, the processor **120** may, in case that the wearing position corresponds to the first position, provide a first wearing guide related to the first position. The first position refers to a wearing position of the wearable device **200** that is above the height of the user's heart and is less affected by venous blood and may correspond to a wearing guide position. The first position may correspond to a case where the angle (e.g., θ) between the wrist direction of the wearable device **200** and the ground has a value exceeding a first reference value (e.g., when the wearable device **200** is worn on the left hand, θ>60°) or less than a second reference value (e.g., when the wearable device **200** is worn on the right hand, θ>60°). The first wearing guide may indicate that oxygen saturation is being measured. Alternatively, the processor **120** may, in case that the wearing position corresponds to the second position, provide a second wearing guide related to the second position.
[0114] The second position may not correspond to the wearing guide position and refer to as a position in which the accuracy of the measurement of oxygen saturation may vary depending on wearing states of the wearable device **200**. The second position may correspond to a case where the angle (e.g., θ) between the wrist direction of the wearable device **200** and the ground has a value falling within a reference range (e.g., a value between the first reference value and the second reference value, −60≤θ≤60°). The second wearing guide may include a guide indicating that the accuracy of measurement of oxygen saturation is reduced, and prompting correct wearing of wearable device **200**. The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the second position, provide the third user interface **520** or the fourth user interface **525** in FIG. 5B.
[0115] Alternatively, the processor **120** may, in case that the wearing position corresponds to the third position, provide a third wearing guide related to the third position. The third position may not correspond to the wearing guide position and correspond to a position that may cause problems with oxygen saturation measurement (e.g., an inaccurate value of oxygen saturation). The third position may correspond to a case where the angle (e.g., θ) between the wrist direction of the wearable device **200** and the ground has a value opposite to the first position (e.g., when the wearable device **200** is worn on the left hand, θ<60°, and when the wearable device **200** is worn on

the right hand, θ>60°). The third wearing guide may include a guide indicating that the measurement of oxygen saturation is impossible, and prompting correct wearing of wearable device **200**. The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the second position, provide the fourth user interface **525** in FIG. **5**B.

[0116] Accordingly, the processor **120** may selectively maintain the oxygen saturation measurement only in case that the wearing position of the wearable device **200** corresponds to the first position or the second position. The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the first position or the second position, perform operations in FIG. **8**. Since oxygen saturation is more likely to be mismeasured when wearing position of the wearable device **200** is the third position, the processor **120** may stop measuring oxygen saturation in case that the wearing position of the wearable device **200** is the third position. The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the third position, not perform operations in FIG. **8**. The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the third position, stop measuring oxygen saturation to prevent degradation of oxygen saturation measurement quality. The processor **120** may prevent a situation that results in low quality oxygen saturation measurement, in advance.

[0117] FIGS. **7**A and **7**B are views illustrating an example of classifying wearing positions according to a wearing hand in a wearable device according to an embodiment.

[0118] FIG. **7**A is a view illustrating an example of classifying wearing positions in case that the wearable device is worn on the left hand.

[0119] Referring to FIG. **7**A, the processor (e.g., the processor **120** in FIG. **1**) of the wearable device (e.g., the electronic device **101** in FIG. **1** or the wearable device **200** in FIG. **2**) according to an embodiment may, in case that the wearing hand of the wearable device **200** is the left hand, classify wearing positions into at least one of a first wearing position **710**, a second wearing position **720**, or a third wearing position **730** according to the tilt of the wearable device **200**. The processor **120** may convert motion data (e.g., x, y, or z) acquired from an inertial sensor (e.g., the sensor module **176** in FIG. **1**) into an Euler angle (e.g., roll, pitch, or yaw), and detect the tilt of the wearable device **200** based on the converted Euler angle.

[0120] The first wearing position **710** refers to a wearing position of the wearable device **200** that is above the height of the user's heart and is less affected by venous blood and may correspond to a wearing guide position. In the first wearing position **710**, the angle (e.g., θ) between the wrist direction **701** of the wearable device **200** and the ground **703** may exceed a determined positive value (e.g., θ>60°).

[0121] The second wearing position **720** may not correspond to the wearing guide position and refer to as a position in which the accuracy of the measurement of oxygen saturation may vary depending on wearing states of the wearable device **200**. In the second wearing position **720**, the angle (e.g., θ) between the wrist direction **701** of the wearable device **200** and the ground **703** may have a value within a determined value range (−60≤θ≤60°). In the second wearing position **720**, the wrist direction of the wearable device **200** and the ground direction may be parallel (e.g., θ=0). The processor **120** may provide a guide prompting the user to correctly wear the wearable device **200** since the accuracy of oxygen saturation measurement is reduced in case that the wearing position of the wearable device **200** corresponds to the second wearing position **720**.

[0122] The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the first wearing position **710** or the second wearing position **720**, perform operations in FIG. **8**.

[0123] The third wearing position **730** may not correspond to the wearing guide position and correspond to a position that may cause problems with oxygen saturation measurement (e.g., an inaccurate value of oxygen saturation). In the third wearing position **730**, the angle (e.g., θ) between the wrist direction **701** of the wearable device **200** and the ground **703** may have a determined negative value (e.g., θ<−60°). Since oxygen saturation is more likely to be mismeasured in the third wearing position **730**, the processor **120** may stop measuring oxygen saturation in case that the wearing position of the wearable device **200** corresponds to the third wearing position **730**. The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the third wearing position **730**, not perform operations in FIG. **8**. The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the third wearing position **730**, stop measuring oxygen saturation to prevent degradation of oxygen saturation measurement quality. The processor **120** may prevent a situation that results in low quality oxygen saturation measurement, in advance.

[0124] FIG. **7**B is a view illustrating an example of classifying wearing positions in case that the wearable device is worn on the right hand.

[0125] Referring to FIG. **7**B, the processor **120** may, in case that the wearing hand of the wearable device **200** is the right hand, classify wearing positions into at least one of a fourth wearing position **750**, a fifth wearing position **760**, or a sixth wearing position **770** according to the tilt of the wearable device **200**. The fourth wearing position **750** refers to a wearing position of the wearable device **200** that is above the height of the user's heart and is less affected by venous blood and may correspond to a wearing guide position. In the fourth wearing position **750**, the angle (e.g., θ) between the wrist direction of the wearable device **200** and the ground may have a negative value (e.g., $\theta > -60°$).

[0126] The fifth wearing position **760** may not correspond to the wearing guide position and refer to as a position in which the accuracy of the measurement of oxygen saturation may vary depending on wearing states of the wearable device **200**. In the fifth wearing position **760**, the angle (e.g., θ) between the wrist direction **701** of the wearable device **200** and the ground **703** may have a value between a specific positive value and a specific negative value ($-60° \leq \theta \leq 60°$). In the fifth wearing position **760**, the wrist direction of the wearable device **200** and the ground direction may be parallel (e.g., $\theta = 0$). The processor **120** may provide a guide prompting the user to correctly wear the wearable device **200** since the accuracy of oxygen saturation measurement is reduced in case that the wearing position of the wearable device **200** corresponds to the fifth wearing position **760**.

[0127] The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the fourth wearing position **750** or the fifth wearing position **760**, perform operations in FIG. **8**.

[0128] The sixth wearing position **770** may not correspond to the wearing guide position and correspond to a position that may cause problems with oxygen saturation measurement (e.g., an inaccurate value of oxygen saturation). In the sixth wearing position **770**, the angle (e.g., θ) between the wrist direction **701** of the wearable device **200** and the ground **703** may have a positive value (e.g., $\theta > 60°$). Since oxygen saturation is more likely to be mismeasured in the sixth wearing position **770**, the processor **120** may stop measuring oxygen saturation in case that the wearing position of the wearable device **200** corresponds to the sixth wearing position **770**. The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the sixth wearing position **770**, not perform operations in FIG. **8**. The processor **120** may, in case that the wearing position of the wearable device **200** corresponds to the sixth wearing position **770**, stop measuring oxygen saturation to prevent degradation of oxygen saturation measurement quality. The processor **120** may prevent a situation that results in low quality oxygen saturation measurement, in advance.

[0129] FIG. **8** is a flowchart **800** illustrating a method of providing a user interface depending on wearing suitability in a wearable device according to an embodiment. FIG. **8** may show operations performed after operation **409** in FIG. **4** or operation **617** in FIG. **6**.

[0130] Referring to FIG. **8**, in operation **801**, a processor (e.g., the processor **120** in FIG. **1**) of the wearable device (e.g., the electronic device **101** in FIG. **1** or the wearable device **200** in FIG. **2**) according to an embodiment may acquire an amplitude of AC and an amplitude of DC from a PPG beat (or PPG signal). The processor **120** may measure oxygen saturation by means of the PPG sensor (e.g., the sensor module **176** in FIG. **1**). The processor **120** may acquire the PPG signal from the PPG sensor and acquire the amplitude of AC and the amplitude of DC from the acquired PPG signal. The PPG signal may be modulated by respiration and thus the processor **120** may determine the wearing suitability of the wearable device **200** by determining whether the modulation caused by respiration is significant. The processor **120** may acquire the amplitude of AC and the amplitude of DC from the acquired PPG signal to determine whether the PPG signal include demodulation caused by respiration.

[0131] According to an embodiment, the processor **120** may use a morphological similarity (Pearson correlation coefficient) between PPG signals. For example, the processor **120** may determine a morphological similarity within an identical wavelength, or determine a morphological similarity between wavelengths. The processor **120** may, in case that the morphological similarity exceeds a determined reference value, perform operation **805**. Alternatively, the processor **120** may compare a waveform template stored in a memory (e.g., the memory **130** in FIG. **1**) and the measured PPG signal to determine a similarity between the waveform template and the measured PPG signal. The waveform template may correspond to an ideal waveform acquired when measuring oxygen saturation using the correct wearing method. The processor **120** may, in case that a similarity between two signals exceeds a determined reference value, perform operation **805**.

[0132] In operation **803**, the processor **120** may determine whether the amplitude of AC exceeds the amplitude of DC. The AC amplitude may be defined as the width of the PPG signal between systole and diastole in a single beat, and the DC amplitude may be defined as the width of the PPG signal between diastole or systole over multiple beats. A length of a window for determining the DC amplitude may be determined to include at least one cycle of modulation by respiration. For example, in case that the wearable device **200** is worn such that the user's body (e.g., wrist) applies moderate pressure to the wearable device **200**, the PPG signal may be acquired primarily as an arterial blood waveform. In this case, in case that the amplitudes of AC and DC are acquired for a determined period (e.g., 6) of a PPG signal period, the AC amplitude may be greater than the DC amplitude (e.g., a ratio of AC amplitude/DC amplitude is high) because a change amount in DC is not large. The determined period may be longer than a respiration cycle. However, in case that the wearable device **200** is not worn such that the user's body (e.g., wrist) applies moderate pressure to the wearable device **200** (e.g., in case of loosely wearing), a venous blood waveform may be introduced into the PPG signal by respiration. In this case, in case that the amplitudes of AC and DC are acquired for a determined period of the PPG signal period, the AC amplitude may be less than the DC amplitude (e.g., a ratio of AC amplitude/DC amplitude is low) because a change amount in DC is large (e.g., a multiple of AC).

[0133] The processor **120** may, in case that the AC amplitude exceeds the DC amplitude, perform operation **805**, and in case that the AC amplitude is less than or equal to the DC amplitude, perform operation **821**.

[0134] In case that the AC amplitude exceeds the DC amplitude, in operation **805**, the processor **120** may determine it corresponds to wearing suitability. The wearing suitability may refer to a state where the wearable device **200** is worn such that the user's body applies moderate pressure to the wearable device **200**. That is, the wearing suitability may refer to a state where the user is correctly wearing the wearable device **200** when oxygen saturation is measured. The wearing suitability may refer to a state where a degree of contact between the user's body and the wearable device **200** is strong, such that there is little (or no) space (or gap) between the user's body and the wearable device **200**.

[0135] In operation **807**, the processor **120** may maintain oxygen saturation measurement. The processor **120** may, in case that the wearing state of the wearable device **200** corresponds to the wearing suitability, continuously measure oxygen saturation.

[0136] In operation **809**, the processor **120** may determine whether the oxygen saturation exceeds a reference value. The oxygen saturation may be used to measure an amount of oxygen in the blood so as to indirectly determine whether oxygen is being adequately delivered to the respiratory system of the body. For example, an oxygen saturation level of 95% or higher is considered normal in cases without respiratory problems. Accordingly, the reference value may be configured to a value (or range of values, (e.g., 90% to 95%)) that is determined to be normal. Alternatively, the reference value may be configured to be changed by the user. For example, in case that the user has a respiratory disorder, the reference value may be configured slightly lower. In case that the user manually configures the reference value, the reference value may be stored in a memory (e.g., the memory **130** of FIG. **1**) of the wearable device **200**. However, a lower bound on the reference value (e.g., 80% or more) may be established to prevent excessive mismeasurement.

[0137] The processor **120** may, in case that the oxygen saturation exceeds the reference value, perform operation **811**, and in case that the oxygen saturation is equal to or less than the reference value, perform operation **813**.

[0138] In case that the oxygen saturation exceeds the reference value (e.g., the quality of the oxygen saturation is high), in operation **811**, the processor **120** may provide an oxygen saturation measurement value. The processor **120** may display the oxygen saturation value (or figure) through a display (e.g., the display module **160** in FIG. **1**, or the display **220** in FIG. **2**). Alternatively, the processor **120** may output the oxygen saturation value as a voice through a speaker (e.g., the sound output module **155** in FIG. **1** or the audio module **205** or **208** in FIG. **2**) of the wearable device **200**. The processor **120** may, in case that the oxygen saturation exceeds the reference value, provide the fifth user interface **530** of FIG. **5**C.

[0139] In case that the oxygen saturation is less than or equal to the reference value (e.g., the quality of oxygen saturation is low), in operation **811**, the processor **120** may provide a warning notification. The processor **120** may display the oxygen saturation measurement value (or figure) and the warning notification (e.g., it may be inaccurate if the wearing hand is down toward the ground) through the display **220**, or may output same as a voice through the audio modules **205** or **208**. The processor **120** may, in case that the oxygen saturation is less than or equal to the reference value, provide the sixth user interface **535** of FIG. **5**C.

[0140] In case that the AC amplitude is less than or equal to the DC amplitude, in operation **821**, the processor **120** may determine it corresponds to wearing unsuitability. The wearing unsuitability may refer to a state where the wearable device **200** is not worn such that the user's body applies moderate pressure to the wearable device **200**. That is, the wearing unsuitability may refer to a state where the user is not correctly wearing the wearable device **200** when oxygen saturation is measured. For example, the wearing unsuitability may refer to a state where a degree of contact between the user's body and the wearable device **200** is weak, such that there is wide space between the user's body and the wearable device **200**.

[0141] According to an embodiment, the processor **120** may use an algorithm without an oxygen saturation output value to determine a signal quality of the oxygen saturation through a coverage within a measurement window section. For example, in case that the oxygen saturation value is output at a frequency of less than 10% within a configured period (e.g., 5 seconds), the processor **120** may determine that it corresponds to wearing unsuitability. In case that the oxygen saturation output value does not exist, the processor **120** may determine the signal quality of the oxygen saturation based on whether the coverage is high or low.

[0142] In operation **823**, the processor **120** may stop oxygen saturation measurement. The processor **120** may, in case that the wearing state of the wearable device **200** corresponds to the wearing unsuitability, stop the oxygen saturation measurement. Because the measured oxygen saturation includes a venous blood waveform due to a wearing defect of the wearable device **200**, the processor **120** may determine that the signal quality of the oxygen saturation is low (e.g., difficulty providing an accurate oxygen saturation value) and no longer measure the oxygen saturation.

[0143] In operation **825**, the processor **120** may determine whether interruption in the oxygen saturation measurement is a first-time occurrence. Although it is described below as providing different measurement guides depending on whether the interruption in oxygen saturation measurement is a first occurrence, it is possible to provide different measurement guides depending on whether the interruption in oxygen saturation measurement has occurred more than or equal to a designated number of times (e.g., three times). This is merely one possible implementation, and the present disclosure is not limited thereto.

[0144] In case that the interruption in oxygen saturation measurement is the first occurrence, in operation **827**, the processor **120** may provide a first measurement guide. The first measurement guide may include a message, such as "Tighten the watch strap" or "Move the wearable device up toward the body (e.g., below or above the elbow)." The first measurement guide may include at least one of text, an image, audio, or a video. The processor **120** may, in case that the interruption in oxygen saturation measurement is the first occurrence, provide the seventh user interface **540** of FIG. **5**D. The processor **120** may display the first measurement guide through the display **220** or output the first measurement guide using a voice through the audio module **205** or **208**. The processor **120** may perform operation **827** and then return to operation **401**.

[0145] In case that the interruption in oxygen saturation measurement is not the first occurrence (e.g., multiple occurrences), in operation **829**, the processor **120** may provide the second measurement guide. The second measurement guide may be different from the first measurement guide. The second measurement guide may include a message, such as "Remove and reattach the wearable device" or "Switch the wearing hand (e.g., guide wearing on the opposite hand)." The second measurement guide may include at least one of text, an image, audio, or a video. The processor **120** may, in case that the interruption in oxygen saturation measurement is not the first occurrence, provide the eighth user interface **545** of FIG. **5**D. The processor **120** may display the second measurement guide through the display **220** or output the second measurement guide using a voice through the audio module **205** or **208**. The processor **120** may perform operation **829** and then return to operation **401**.

[0146] FIGS. **9**A and **9**B are views illustrating an example of classifying wearing suitability of a wearable device according to an embodiment.

[0147] FIG. **9**A is a view illustrating an example of a PPG signal acquired according to a wearing state of the wearable device.

[0148] Referring to FIG. **9**A, a processor (e.g., the processor **120** in FIG. **1**) of the wearable device (e.g., the electronic device **101** in FIG. **1** or the wearable device **200** in FIG. **2**) according to an embodiment may acquire an amplitude of AC and an amplitude of DC from a PPG beat (or PPG signal). The AC amplitude may be defined as the width of the PPG signal between systole and diastole in a single beat, and the DC amplitude may be defined as the width of the PPG signal between diastole or systole over multiple beats. A length of a window for determining the DC amplitude may be determined to include at least one cycle of modulation by respiration. For example, in case that the user is wearing the wearable device **200** correctly (e.g., wearing suitability) when measuring oxygen saturation, the processor **120** may acquire a first PPG signal **910**. In the first PPG signal **910**, the AC amplitude **901** may be greater than the DC amplitude **903** (e.g., a ratio of AC amplitude/DC amplitude is high). However, in case that the user is wearing the wearable device **200** incorrectly (e.g., wearing unsuitability) when measuring oxygen saturation, the processor **120** may acquire a second PPG signal **920**. In the second PPG signal **920**, the AC amplitude **921** may be less than the DC amplitude **923** (e.g., a ratio of AC amplitude/DC amplitude is low).

[0149] FIG. **9**B is a graph illustrating an example of a PPG signal acquired according to a wearing state of the wearable device.

[0150] Referring to FIG. **9**B, a first signal graph **930** depicts a PPG signal acquired according to the wearing state of the wearable device **200**. The first PPG signal **931** may be acquired when infrared is used for the light-emitting unit, and the second PPG signal **933** may be acquired when RED is used for the light-emitting unit. The first PPG signal **931** and the second PPG signal **933** may show similar patterns according to the wearing state of the wearable device **200**. For example, in case that the user is wearing the wearable device **200** correctly, PPG signals, such as a first signal period **943**, a second signal period **945**, and a third signal period **947**, may be acquired. In the first signal period **943**, the second signal period **945**, and the third signal period **947**, the AC amplitude may be greater than the DC amplitude (e.g., the ratio of AC amplitude/DC amplitude is high). In case that the user is not wearing the wearable device **200** correctly, a PPG signal, such as a fourth signal period **935**, may be acquired. In the fourth signal period **935**, the AC amplitude may be less than the DC amplitude (e.g., the ratio of AC amplitude/DC amplitude is low).

[0151] In addition, a second graph **950** may depict a PPG signal acquired when the user is wearing the wearable device **200** correctly. In case that the amplitudes of AC and DC are acquired from the PPG signal as shown in the second graph **950**, the AC amplitude may be greater than the DC because a change amount in DC is not large. A third graph **960** may depict a PPG signal acquired when the user is not wearing the wearable device **200** correctly. In case that the amplitudes of AC and DC are acquired from the PPG signal as shown in the third graph **960**, the AC amplitude may be less than the DC amplitude because a change amount in DC is large.

[0152] FIG. **10** is a view illustrating an example of providing a guide associated with oxygen saturation measurement in a wearable device according to an embodiment.

[0153] Referring to FIG. **10**, a processor (e.g., the processor **120** in FIG. **1**) of the wearable device (e.g., the electronic device **101** in FIG. **1** or the wearable device **200** in FIG. **2**) according to an embodiment may, in case that a quality of a signal is not good, additionally provide at least one of a first measurement guide **1010** to a fourth measurement guide **1070**. For example, for higher signal quality, the processor **120** may provide the first measurement guide **1010** to guide the user to place the wearable device **200** up to the eye level. The processor **120** may provide the second measurement guide **1030** to guide the user to place the wearable device **200** above the heart level **1001** to reduce an effect of venous blood. For higher signal quality, the processor **120** may provide the third measurement guide **1050** or the fourth measurement guide **1070** to guide the user to support the wearable device **200** using the opposite hand (e.g., a hand that is not wearing the wearable device **200**) to avoid shaking. The third measurement guide **1050** may show a support point (e.g., an elbow **1051**) to minimize shaking of the wearable device **200** using the opposite hand. The fourth measurement guide **1070** may show a support point (e.g., the back of the other hand **1057**) to minimize shaking of the wearable device **200** using the opposite hand.

[0154] According to an embodiment, an operation method of a wearable device **200** may include an operation of, in response to a request for measurement of oxygen saturation, acquiring motion data from an inertial sensor (e.g., sensor module **176**) included in the wearable device, an operation of, in case that it is determined based on the motion data that there is no motion in a wearable device, determining a wearing hand on which the wearable device is worn, based on the motion data, an operation of detecting a tilt of the wearable device while measuring the oxygen saturation by means of a PPG sensor (e.g., sensor module **176**) included in the wearable device, an operation of classifying wearing positions of the wearable device based on the wearing hand and the tilt, and an operation of providing different user interfaces according to the wearing positions.

[0155] The method may further include an operation of, in case that the wearing hand corresponds to the left hand, applying a left hand tilt detection algorithm, an operation of determining whether an angle between a wrist direction of the wearable device and the ground exceeds a first reference value according to the left hand tilt detection algorithm, and an operation of classifying the wearing position of the wearable device according to a result of the determination.

[0156] The method may further include an operation of, in case that the wearing hand corresponds to the right hand, applying a right hand tilt detection algorithm, an operation of determining whether the angle between the wrist direction of the wearable device and the ground is less than a second value according to the right hand tilt detection algorithm, and an operation of classifying the wearing position of the wearable device according to a result of the determination.

[0157] The method may perform one of an operation of, in case that the wearing position corresponds to a first position, providing a first user interface associated with the first position, an operation of, in case that the wearing position corresponds to a second position, providing a second user interface associated with the second position, and an operation of, in case that the wearing position corresponds to a third position, providing a third user interface associated with the third position, wherein the first position corresponds to a case where the angle between the wrist direction of the wearable device and the ground having a value exceeding a first reference value or less than a second reference value, the second position corresponds to a case where the angle between the wrist direction of the wearable device and the ground has a value between the first reference value and the second reference value, and the third position corresponds to a case where the angle between the wrist direction of the wearable device and the ground has a value opposite to the first position.

[0158] The method may further include an operation of, in case that the wearing position corresponds to the first position or the second position, maintaining oxygen saturation measurement and an operation of, in case that the wearing position corresponds to the third position, stopping oxygen saturation measurement.

[0159] The method may include an operation of acquiring an amplitude of AC and an amplitude of DC from a PPG signal acquired from the PPG sensor, an operation of, in case that the amplitude of the AC exceeds the amplitude of the DC, determining that a wearing state of the wearable device is suitable, and an operation of, in case that the amplitude of the AC is equal to or less than the amplitude of the DC, determining that the wearing state of the wearable device is unsuitable.

[0160] The method may include an operation of, when it is determined that the wearing state of the wearable device is suitable, maintaining the oxygen saturation measurement, an operation of determining whether the measured oxygen saturation exceeds a reference value, an operation of, in case that the measured oxygen saturation exceeds the reference value, providing an oxygen saturation measurement value, and in case that the measured oxygen saturation is equal to or less than the reference value, providing the oxygen saturation measurement value and a warning notification, an operation of, when it is determined that the wearing state of the wearable device is unsuitable, stopping the oxygen saturation measurement, and an operation of, in case that the number of oxygen saturation measurement stops corresponds to a designated number of times, providing a first measurement guide, and in case that the number of oxygen saturation measurement stops does not correspond to the designated number of times, providing a second measurement guide.

[0161] The embodiments disclosed in the specification and the drawings are merely presented as specific examples to easily explain the technical features and help understanding of the disclosure and are not intended to limit the scope of the disclosure. Therefore, the scope of the disclosure should be construed as encompassing all changes or modifications derived from the technical ideas of the disclosure in addition to the embodiments disclosed herein.

Claims

1. A wearable device comprising: an inertial sensor; a photoplethysmography (PPG) sensor; a display module; a processor operatively coupled to at least one of the inertial sensor, the PPG sensor, and the display module; and a memory connected electrically to the processor and configured to store instructions executable by the processor, wherein the instructions, when executed by the processor, cause the wearable device to: in response to a request for measurement of oxygen saturation, acquire motion data from the inertial sensor; responsive to a determination, based on the motion data, that there is no motion of the wearable device, determine a wearing hand on which the wearable device is worn, based on the motion data; detect a tilt of the wearable device during the measurement of oxygen saturation by means of the PPG sensor; classify wearing positions of the wearable device, based on the wearing hand and the tilt; and provide different user interfaces according to the wearing positions.

2. The wearable device of claim 1, wherein the instructions are configured to, when executed by the processor, cause the wearable device to: in case that the wearing hand corresponds to a left hand, apply a left-hand tilt detection algorithm; and in case that the wearing hand corresponds to a right hand, apply a right-hand tilt detection algorithm.

3. The wearable device of claim 2, wherein the instructions are configured to, when executed by the processor, cause the wearable device to: determine whether an angle between a wrist direction of the wearable device and a ground exceeds a first reference value according to the left-hand tilt detection algorithm; and classify the wearing positions of the wearable device according to a result of the determination.

4. The wearable device of claim 2, wherein the instructions are configured to, when executed by the processor, cause the wearable device to: determine whether an angle between a wrist direction of the wearable device and a ground is less than a second reference value according to the right-hand tilt detection algorithm; and classify the wearing positions of the wearable device according to a result of the determination.

5. The wearable device of claim 1, wherein the instructions are configured to, when executed by the processor, cause the wearable device to: in case that a wearing position corresponds to a first position, provide a first user interface associated with the first position; in case that the wearing position corresponds to a second position, provide a second user interface associated with the second position; and in case that the wearing position corresponds to a third position, provide a third user interface associated with the third position, wherein the first position corresponds to a case where an angle between a wrist direction of the wearable device and a ground has a first value exceeding a first reference value or less than a second reference value, wherein the second position corresponds to a case where the angle between the wrist direction of the wearable device and the ground has a second value between the first reference value and the second reference value, and wherein the third position corresponds to a case where the angle between the wrist direction of the wearable device and the ground has a third value opposite to the first position.

6. The wearable device of claim 5, wherein the instructions are configured to, when executed by the processor, cause the wearable device to: in case that the wearing position corresponds to the first position or the second position, maintain the measurement of oxygen saturation; and in case that the wearing position corresponds to the third position, stop oxygen saturation measurement.

7. The wearable device of claim 1, wherein the instructions are configured to, when executed by the processor, cause the wearable device to: acquire an amplitude of alternating current (AC) and an amplitude of direct current (DC) from a PPG signal acquired from the PPG sensor; in case that the amplitude of the AC exceeds the amplitude of the DC, determine that a wearing state of the wearable device is appropriate; and in case that the amplitude of the AC is equal to or less than the amplitude of the DC, determine that the wearing state of the wearable device is inappropriate.

8. The wearable device of claim 7, wherein the instructions are configured to, when executed by the processor, cause the wearable device to, responsive to a determination that the wearing state of the wearable device is appropriate, maintain the measurement of oxygen saturation.

9. The wearable device of claim 8, wherein the instructions are configured to, when executed by the processor, cause the wearable device to: determine whether a measured oxygen saturation exceeds a reference value; in case that the measured oxygen saturation exceeds the reference value, provide an oxygen saturation measurement value; and in case that the measured oxygen saturation is equal to or less than the reference value, provide the oxygen saturation measurement value and a warning notification.

10. The wearable device of claim 7, wherein the instructions are configured to, when executed by

the processor, cause the wearable device to, responsive to a determination that the wearing state of the wearable device is inappropriate, stop the measurement of oxygen saturation.
11. The wearable device of claim 10, wherein the instructions are configured to, when executed by the processor, cause the wearable device to: in case that a number of stops of the measurement of oxygen saturation corresponds to a designated number of times, provide a first measurement guide; and in case that the number of stops of the measurement of oxygen saturation does not correspond to the designated number of times, provide a second measurement guide.
12. The wearable device of claim 11, wherein the instructions are configured to, when executed by the processor, cause the wearable device to, after providing the first measurement guide or the second measurement guide, determine whether there is no motion of the wearable device.
13. The wearable device of claim 1, wherein the instructions are configured to, when executed by the processor, cause the wearable device to: after the oxygen saturation measurement is requested, in case that a number of detections of the motion of the wearable device corresponds to a designated number of detections, provide a first guide; and after the oxygen saturation measurement is requested, in case that a number of detections of the motion of the wearable device does not correspond to the designated number of detections, provide a second guide.
14. An operation method of a wearable device, the method comprising: in response to a request for measurement of oxygen saturation, acquiring motion data from an inertial sensor included in the wearable device; if it is determined, based on the motion data, that there is no motion of the wearable device, determining a wearing hand on which the wearable device is worn, based on the motion data; detecting a tilt of the wearable device during the measurement of oxygen saturation by means of a photoplethysmography (PPG) sensor included in the wearable device; classifying wearing positions of the wearable device, based on the wearing hand and the tilt; and providing different user interfaces according to the wearing positions.
15. The method of claim 14, further comprising: in case that the wearing hand corresponds to a left hand, applying a left-hand tilt detection algorithm; determining whether an angle between a wrist direction of the wearable device and a ground exceeds a first reference value according to the left-hand tilt detection algorithm; and classifying the wearing positions of the wearable device according to a result of the determination
16. The method of claim 14, further comprising: in case that the wearing hand corresponds to a right hand, applying a right-hand tilt detection algorithm; determining whether the angle between the wrist direction of the wearable device and the ground is less than a second reference value according to the right-hand tilt detection algorithm; and classifying the wearing positions of the wearable device according to a result of the determination.
17. The method of claim 14, further comprising: in case that a wearing position corresponds to a first position, providing a first user interface associated with the first position; in case that the wearing position corresponds to a second position, providing a second user interface associated with the second position, and in case that the wearing position corresponds to a third position, providing a third user interface associated with the third position, wherein the first position corresponds to a case where an angle between the wrist direction of the wearable device and a ground having a first value exceeding a first reference value or less than a second reference value, wherein the second position corresponds to a case where the angle between the wrist direction of the wearable device and the ground has a second value between the first reference value and the second reference value, and wherein the third position corresponds to a case where the angle between the wrist direction of the wearable device and the ground has a third value opposite to the first position.
18. The method of claim 17, further comprising: in case that the wearing position corresponds to the first position or the second position, maintaining oxygen saturation measurement and an operation of, and in case that the wearing position corresponds to the third position, stopping oxygen saturation measurement.

19. The method of claim 14, further comprising: acquiring an amplitude of alternating current (AC) and an amplitude of direct current (DC) from a PPG signal acquired from the PPG sensor; in case that the amplitude of the AC exceeds the amplitude of the DC, determining that a wearing state of the wearable device is suitable; and in case that the amplitude of the AC is equal to or less than the amplitude of the DC, determining that the wearing state of the wearable device is unsuitable.
20. The method of claim 14, further comprising: when it is determined that the wearing state of the wearable device is suitable, maintaining the oxygen saturation measurement, an operation of determining whether the measured oxygen saturation exceeds a reference value; in case that the measured oxygen saturation exceeds the reference value, providing an oxygen saturation measurement value; in case that the measured oxygen saturation is equal to or less than the reference value, providing the oxygen saturation measurement value and a warning notification when it is determined that the wearing state of the wearable device is unsuitable, stopping the oxygen saturation measurement; in case that the number of oxygen saturation measurement stops corresponds to a designated number of times, providing a first measurement guide; and in case that the number of oxygen saturation measurement stops does not correspond to the designated number of times, providing a second measurement guide.